

Project Proposal and Strategic Methodology: Project AAPEMS

Executive Summary

This document outlines our comprehensive research proposal for our (MTP). Building upon the initial exploratory codebase (which utilized a standard RAG setup with all-mpnet-base-v2 embeddings, Pinecone vector search, and single-pass LLM prompting), we have identified that we simply cannot stick to static, single-pass classification if we want a solid thesis.

To tackle the limitations of existing baselines—such as the lack of deep clinical reasoning, unfaithful explanations, and failure to handle real-world complexities—we propose **Project AAPEMS (Agentic AI-based Patient Eligibility Matching System)**. This framework transforms clinical trial matching from a one-shot guess into an iterative, self-improving, multi-agent reasoning workflow.

1. Research Gap and Motivation

When we look closely at prior literature (including TrialGPT, SIGIR benchmarks, and studies on NLI annotation artifacts and LLM faithfulness), it is clear there are major gaps in current automated screening systems:

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1. **The Execution vs. Reasoning Gap:** Existing systems just translate criteria into SQL or retrieval queries, but they lack deep clinical reasoning to properly handle temporal conditions, negations, disease–stage requirements, and lab thresholds.
2. **Lack of Self–Correction:** Most models operate merely as one–time binary classifiers. They do not inspect their own mistakes, figure out false positives or false negatives, or refine their reasoning across multiple execution cycles.
3. **Prevalence Shift & Calibration:** Academic evaluations mostly rely on neat, balanced datasets, whereas real–world clinical deployment involves severe class imbalance where truly eligible patients are extremely rare.

2. Proposed Methodology: Project AAPEMS

We are proposing an autonomous, multi–agent framework designed to bridge the gap between automated execution and clinical reasoning across six operational layers:

A. Data Preparation & Ingestion

1. **Datasets:** Grounded firmly in real–world benchmarks, including ClinicalTrials.gov protocols, MIMIC–IV, eICU, and the Syneos NSCLC dataset (which we have structured cleanly into an 80% training set and a 20% test set).
2. **Signal Integration:** Unifies structured Electronic Health Record (EHR) data (like demographics, lab values, and diagnosis codes) together with unstructured clinical notes (like radiology reports and physician observations).

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B. Semantic Parsing & Clinical Reasoning

Free-text clinical trial eligibility criteria are passed through a semantic parsing module. A dedicated **Clinical Reasoning Agent** then interprets complex medical logic—including temporal constraints, negations, treatment-history dependencies, and laboratory thresholds—before any query or matching execution takes place.

C. Structured Representation & Execution

The reasoned clinical logic is converted into structured queries (such as SQL for EHR database cohort retrieval or semantic vector queries via Pinecone using all-mpnet-base-v2 embeddings).

D. Autonomous Feedback and Refinement Loop

Instead of stopping at a single prediction, AAPEMS brings in specialized error-analysis agents:

1. **False Positive Analyzer Agent:** Detects when a patient was incorrectly included due to overlooked exclusion criteria or misunderstood constraints.
2. **False Negative Analyzer Agent:** Identifies valid eligible patients who were unfortunately missed because of overly strict filtering.
3. **Refinement Agent:** Dynamically adjusts reasoning pathways and re-runs the pipeline across multiple iterative cycles (roughly 3 to 5 cycles) until convergence.

E. Advanced Evaluation Suite

4. **Standard Metrics:** Execution Accuracy, Classification Accuracy, Precision, Recall, and F1–Score.
5. **Advanced Research Metrics:** Precision@K, Recall@K, Brier Score, Expected Calibration Error (ECE), and Prevalence–Shift Robustness.
6. **Statistical Rigor:** Application of multi–test correction procedures (such as Holm's step–down procedure) when evaluating multiple model configurations and hypotheses simultaneously.

3. Comparison with Alternative Approaches

We also looked at alternative conservative extensions (like adding basic rule–based checks or static structured extraction). However, those approaches remain heavily restricted to single–pass processing:

1. **Single–Pass Vulnerability:** They totally lack active error–analysis agents to diagnose why a failure happened or trigger targeted prompt/logic refinement.
2. **Lack of Novelty:** Minor tweaks to a basic tutorial–level RAG script simply will not meet the rigorous algorithmic novelty and robustness standards required for an MTP.
3. **Deployment Realism:** By incorporating proper calibration metrics (ECE) and prevalence–shift evaluation, AAPEMS bridges the wide gap between toy academic benchmarks and deployable clinical decision support.

4. Conclusion

Project AAPEMS provides the exact technical rigor, architectural novelty, and methodological depth required to deliver a high-impact MTP. By turning patient matching into a self-improving, multi-agent reasoning framework, we ensure our research addresses real-world clinical complexities. We look forward to your feedback and approval to proceed fully with the implementation.